

PHASE-I PROJECT

BCSE332L – Deep Learning

Energy AI / Battery Management Systems (BMS)

Team No

12

Team Members (Reg Nos)

23BAI0093, 23BAI0157, 23BAI0143

Adaptive Physics-Informed Battery Digital Twin for State-of-Health (SOH) and Remaining Useful Life (RUL) Prediction

Under

CHELLATAMILAN SIR

1. Problem Identification

Core Problem (One Line)

Existing battery health prediction methods either rely on large labeled datasets or computationally intensive physics-based models, making accurate, physically consistent, and real-time estimation of State of Health (SOH) and Remaining Useful Life (RUL) challenging.

2. Problem Description:

Lithium-ion batteries are the primary energy storage devices used in electric vehicles, renewable energy storage systems, drones, consumer electronics, and smart grids. Their performance gradually deteriorates due to repeated charging and discharging cycles, aging, temperature variations, and varying operating conditions. Accurate estimation of **State of Health (SOH)** and **Remaining Useful Life (RUL)** is essential for ensuring battery safety, reliability, predictive maintenance, and efficient energy management. Traditional electrochemical models provide physically meaningful predictions but are computationally intensive and unsuitable for real-time deployment.

In contrast, deep learning models can achieve high prediction accuracy but require large labeled datasets and may produce physically inconsistent predictions that violate battery degradation principles. Recent research has shown that Physics-Informed Neural Networks (PINNs) effectively combine battery physics with deep learning to improve prediction reliability. However, most existing approaches are trained offline and require complete retraining whenever battery degradation characteristics change.

Therefore, this project proposes an **Adaptive Physics-Informed Battery Digital Twin** that integrates Transformer-based temporal feature extraction, Physics-Informed Neural Networks (PINNs), and incremental adaptive learning to provide accurate, physically consistent, and continuously updated prediction of battery SOH and RUL.

3. Relevance to Real-world Scenarios

The proposed framework is highly relevant to Battery Management Systems (BMS) used in electric vehicles, renewable energy storage systems, drones, consumer electronics, and industrial battery storage. Accurate prediction of battery health enables predictive maintenance, improves operational safety, reduces unexpected battery failures, lowers maintenance costs, extends battery lifetime, and supports intelligent energy management. An adaptive Battery Digital Twin can continuously update itself using real operational data, enabling personalized and real-time battery monitoring throughout its lifecycle.

4. Objectives

Objective 1

Develop a Physics-Informed Battery Digital Twin capable of accurately estimating State of Health (SOH) and Remaining Useful Life (RUL) using battery sensor data and electrochemical constraints.

Justification

Traditional deep learning models often ignore battery physics, whereas electrochemical models are computationally expensive. Combining both approaches improves prediction accuracy while maintaining physical consistency.

Objective 2

Develop an adaptive incremental learning framework that continuously updates the Battery Digital Twin using newly acquired operational data without retraining the entire model from scratch.

Justification

Battery degradation evolves continuously because of aging, charging patterns, and environmental conditions. Adaptive learning enables the Digital Twin to remain accurate throughout the battery lifecycle while reducing computational cost.

5. How These Objectives Solve the Problem

The first objective combines battery physics with deep learning to generate accurate and physically meaningful predictions of SOH and RUL. The second objective enables continual adaptation of the Digital Twin using newly collected battery data, eliminating expensive retraining and improving long-term prediction reliability. Together, these objectives provide an intelligent and adaptive framework suitable for real-time Battery Management Systems.

6. Existing Models

Strengths

- High prediction accuracy using deep learning
- Effective temporal feature extraction using Transformer architectures
- Physics-informed learning improves prediction consistency
- Supports simultaneous estimation of SOH and RUL
- Captures complex battery degradation behaviour

Limitations

- Most models are trained offline.
- Require complete retraining as battery behaviour changes.
- Limited personalization for individual batteries.
- Depend on large labelled datasets.
- High computational complexity for real-time deployment.

7. Novel Observations / Data Trends

Recent studies indicate increasing adoption of Physics-Informed Neural Networks (PINNs) integrated with Transformer architectures for battery health prediction. However, adaptive Battery Digital Twins capable of continual incremental learning for personalized real-time battery monitoring remain an active research direction. Integrating adaptive learning with PINNs can significantly improve long-term prediction accuracy while reducing retraining cost.

8. Deep Learning Techniques Used

- Physics-Informed Neural Networks (PINNs)
- Transformer Encoder
- Feedforward Deep Neural Network (DNN)
- Multi-task Learning
- Incremental / Continual Learning
- Battery Digital Twin Framework

9. Standard Dataset

NASA Lithium-Ion Battery Aging Dataset

Dataset Source

NASA Prognostics Center of Excellence (PCoE)

[Link: https://www.nasa.gov/intelligent-systems-division/discovery-and-systems-health/pcoe/pcoe-data-set-repository/](https://www.nasa.gov/intelligent-systems-division/discovery-and-systems-health/pcoe/pcoe-data-set-repository/)

10. Base Reference Paper

Physics-informed neural network for co-estimation of state of health, remaining useful life, and short-term degradation path in lithium-ion batteries

Journal

Applied Energy (2025)

DOI

[Link: https://doi.org/10.1016/j.apenergy.2025.126427](https://doi.org/10.1016/j.apenergy.2025.126427)

ANNEXURE

Literature Review & Comparative Analysis

S.No	Paper Title	Authors	Journal	Problem Addressed	Methodology	Dataset Used	Strengths	Limitations	Research Gap
1	Physics-informed neural networks for battery modeling and state estimation: A methodological review	G. Yu et al.	Renewable & Sustainable Energy Reviews	Reviews PINN applications for battery modelling and state estimation.	Comprehensive review of PINN methodologies.	-	Summarizes recent developments and future directions.	Does not propose a new prediction model.	Need for adaptive and deployable Battery Digital Twins.
2	Physics-informed neural network for co-estimation of SOH, RUL and degradation path in lithium-ion batteries	L. Yang et al.	Applied Energy	Joint estimation of SOH, RUL and degradation path.	Improved Transformer + PINN + Multi-task Learning.	NASA PCoE Battery Dataset	High prediction accuracy and multi-task capability.	Offline training requiring retraining.	Lack of continual adaptation for changing battery behaviour.
3	Physical Informed Neural Network for SOH Estimation of Lithium-ion Battery with Electrochemical Mechanism	H. Ma et al.	Energy	Electrochemical SOH estimation.	Electrochemical Mechanism PINN (EM-PINN).	XJTU, MIT and HUST datasets	Incorporates electrochemical degradation mechanisms.	Limited to SOH estimation.	No Digital Twin or adaptive learning.
4	A Cycle-Aware and Physics-Informed Framework for Battery Remaining Useful Life Prediction	—	Scientific Reports	RUL prediction using cycle information.	Cycle-aware Physics-informed Framework.	-	Improved degradation modelling.	Offline training.	No adaptive Digital Twin.
5	Physics-Informed Neural Network and Momentum Contrastive Learning for Battery State of Health Estimation	—	Complex & Intelligent Systems	SOH estimation with limited labelled data.	PINN + Momentum Contrastive Learning.	NASA Lithium-ion Battery Dataset	Better feature representation and reduced label dependency.	Train-then-deploy paradigm.	No online adaptation after deployment.

References

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<https://www.sciencedirect.com/science/article/abs/pii/S1364032126005253>
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5. *Physics-Informed Neural Network and Momentum Contrastive Learning for Battery State of Health Estimation*. Complex & Intelligent Systems.
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https://www.researchgate.net/publication/398786756_Physics-informed_neural_network_and_momentum_contrastive_learning_for_battery_state_of_health_estimation